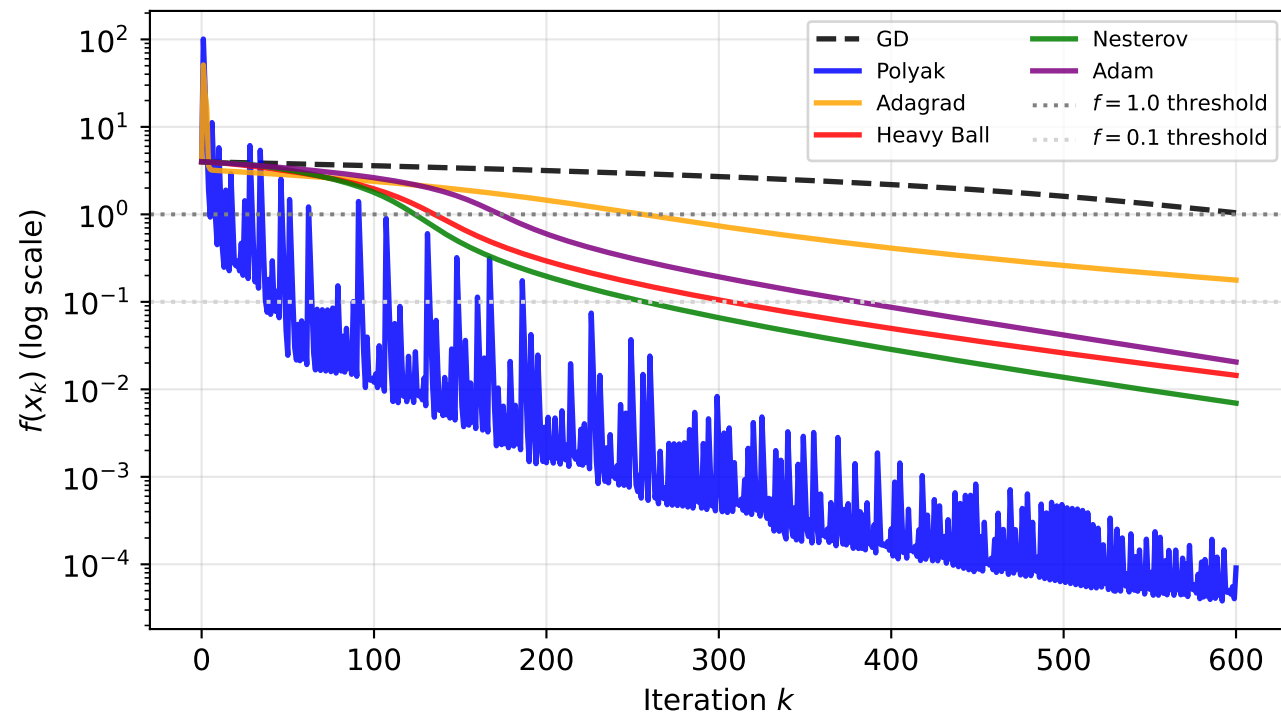


Long-Horizon Convergence on Rosenbrock — All Methods

Iterations to Reach Target $f < \text{threshold}$
(Benchmark C)

All Q1\&Q2 Methods on Benchmark C (600 iterations)



Method	$f < 5.0$	$f < 2.0$	$f < 1.0$	$f < 0.5$	$f < 0.1$
GD	0	434	>600	>600	>600
Polyak	0	4	5	9	38
Adagrad	0	142	256	364	>600
Heavy Ball	0	99	136	167	307
Nesterov	0	94	124	149	257
Adam	0	133	174	212	381